Research Paper

A Modular Framework for Vision-Based 6-DoF Pose Estimation and Motion Analysis Using Common Geometric Representations

Bhaumit Joshi

Abstract

Vision-based motion analysis has become an important tool in engineering measurement, robotics, biomechanics, and industrial inspection. Although numerous computer vision algorithms exist for feature tracking, camera calibration, pose estimation, and three-dimensional reconstruction, most available software systems are designed around fixed computational workflows that tightly couple individual algorithms with specific measurement procedures. This limits their adaptability to diverse engineering applications requiring different sources of geometric information or alternative pose estimation strategies.

This paper presents a modular computer vision framework that unifies multiple engineering workflows through common geometric representations. The proposed framework separates geometric information acquisition from its subsequent utilization by providing interchangeable coordinate acquisition methods, including direct engineering measurement, CAD-assisted coordinate generation, and stereo reconstruction, together with multiple pose estimation workflows based on SolvePnP and Kabsch rigid-body registration. Standardized engineering representations enable downstream motion analysis, signal processing, visualization, and reporting modules to operate independently of the selected computational methodology.

The framework was implemented as an integrated desktop application using C#, WPF, OpenCvSharp, HelixToolkit, and Math.NET. Experimental validation was performed using real engineering datasets. Camera calibration, stereo reconstruction, and monocular pose estimation were evaluated using independently obtained reference measurements, demonstrating reliable recovery of rigid-body motion suitable for engineering analysis. The modular architecture further facilitates future integration of additional computer vision methodologies without modification of the downstream computational workflow.

The proposed framework demonstrates that established computer vision algorithms can be integrated within a unified engineering architecture that improves flexibility, extensibility, and practical applicability for quantitative vision-based motion analysis.

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1. Introduction

Vision-based motion analysis has become an important tool in engineering disciplines including robotics, biomechanics, manufacturing, structural monitoring, industrial metrology, and human movement analysis **[1–4]**. Advances in digital imaging, camera calibration, feature tracking, and pose estimation have enabled non-contact measurement of rigid-body motion with increasing accuracy and flexibility **[1,2,5,6]**. Compared with traditional contact-based measurement techniques, vision-based approaches provide the ability to analyse complex motions while preserving the natural behaviour of the observed system and reducing instrumentation requirements **[2–4]**.

Numerous computational techniques have been developed to support individual stages of the vision-based motion analysis process. Camera calibration methods establish the geometric relationship between image and object space **[5,7]**, feature tracking algorithms determine the temporal location of visual landmarks **[8,9]**, pose estimation algorithms recover rigid-body transformations **[10,11]**, and stereo vision techniques enable three-dimensional reconstruction from synchronized image observations **[2,12]**. These methods are well established, and robust implementations are available through widely used computer vision libraries such as OpenCV **[13]**.

Despite the maturity of these individual algorithms, practical engineering software often remains tightly coupled to specific computational workflows. Many systems assume that object-space coordinates originate from a single source, employ only one pose estimation methodology, or integrate processing stages in a manner that limits adaptability to different engineering measurement scenarios. Consequently, extending such systems to incorporate alternative coordinate acquisition methods, additional pose estimation strategies, or new analysis procedures frequently requires substantial modification of the underlying software architecture.

For many engineering applications, however, rigid-body motion analysis should be viewed as a sequence of engineering tasks rather than a fixed sequence of computational algorithms. The acquisition of geometric information, estimation of rigid-body pose, computation of motion parameters, signal processing, visualization, and engineering interpretation represent distinct functional stages that may each be realized through multiple computational methodologies. A software framework capable of separating these engineering tasks from their algorithmic implementation can therefore provide greater flexibility while improving maintainability and extensibility.

This paper presents a modular framework for vision-based 6-DoF pose estimation and motion analysis using common geometric representations. The proposed framework unifies multiple coordinate acquisition workflows, including direct engineering measurement, CAD-assisted coordinate generation, and stereo reconstruction, together with multiple pose estimation methodologies based on SolvePnP and Kabsch rigid-body registration. By expressing the outputs of these computational processes through standardized engineering representations, downstream motion analysis, signal processing, visualization, and reporting modules remain independent of the selected computational methodology.

The proposed framework was implemented as an integrated desktop application and evaluated using representative engineering measurement scenarios. Experimental validation includes intrinsic and stereo camera calibration, object-space coordinate generation, monocular pose estimation using independently obtained reference measurements, and complete motion analysis workflows. The results demonstrate that the proposed architecture successfully integrates established computer vision techniques within a unified and extensible engineering framework suitable for quantitative vision-based motion analysis.

The principal contributions of this work are as follows:

1. A modular computer vision framework that separates engineering functionality from algorithm-specific implementation through standardized geometric representations.
2. A unified architecture supporting multiple coordinate acquisition and pose estimation workflows within a common computational environment.
3. An integrated software realization combining camera calibration, feature tracking, object-space coordinate generation, pose estimation, motion analysis, signal processing, and visualization.
4. Experimental validation demonstrating the practical applicability of the proposed framework using real engineering measurements.

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2. Related Work

Vision-based motion analysis has been extensively investigated across a wide range of engineering and scientific applications. Research has produced mature computational techniques for camera calibration, feature tracking, three-dimensional reconstruction, pose estimation, and motion analysis, enabling accurate non-contact measurement of rigid-body motion using digital imaging systems **[1–4]**. Widely adopted computer vision libraries, particularly OpenCV, provide robust implementations of many of these algorithms and have significantly accelerated the development of practical vision-based measurement systems **[13]**.

Camera calibration forms the foundation of vision-based measurement by establishing the geometric relationship between image observations and object space. The flexible calibration technique proposed by **Zhang** has become one of the most widely adopted methods for estimating intrinsic camera parameters and lens distortion **[5]**, while earlier work by **Tsai** established one of the first practical calibration procedures for machine vision metrology **[6]**. Stereo calibration further extends this process by estimating the relative orientation and translation between multiple cameras, enabling image rectification and metric three-dimensional reconstruction **[2,12,14]**.

Feature tracking has likewise received considerable attention. Classical approaches such as the **Lucas–Kanade optical flow** algorithm remain widely used because of their computational efficiency and robustness for sparse feature tracking **[8]**, while the **Shi–Tomasi** feature selection method provides reliable interest point detection for tracking applications **[9]**. More recent developments include correlation-filter-based tracking algorithms such as **CSR-DCF**, which improve robustness under changes in appearance and partial occlusion **[15]**. Together, these approaches provide mature solutions for maintaining temporal correspondence throughout image sequences.

Rigid-body pose estimation represents another fundamental component of vision-based measurement systems. Monocular approaches based on the **Perspective-n-Point (PnP)** problem estimate object pose from known object-space coordinates and corresponding image observations **[10,11]**, whereas stereo vision reconstructs three-dimensional feature coordinates through triangulation using calibrated camera geometry **[2,12,14]**. When three-dimensional point correspondences are available, rigid-body registration algorithms such as the **Kabsch algorithm** provide efficient estimation of the optimal rigid-body transformation between successive point sets **[16]**, while the methods proposed by **Arun et al.** and **Horn** provide alternative least-squares solutions to the absolute orientation problem **[17,18]**.

Although these individual algorithms are well established and robust software implementations are available through libraries such as **OpenCV [13]**, many engineering software systems continue to organize them into fixed computational pipelines tailored to specific applications. Such systems commonly assume predefined sources of object-space coordinates, employ only a single pose estimation methodology, or tightly couple downstream motion analysis to particular computational workflows. These design choices can reduce flexibility when alternative coordinate acquisition methods or pose estimation strategies are required.

The work presented in this paper differs from previous approaches by focusing on **software architecture rather than algorithmic innovation**. Instead of proposing new computer vision algorithms, the proposed framework integrates established methodologies within a modular engineering architecture based on standardized geometric representations. This organization allows multiple coordinate acquisition methods, pose estimation workflows, motion analysis procedures, and visualization components to coexist within a unified software environment while remaining computationally independent through common engineering data structures.

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3. Proposed Framework

3.1 Framework Overview

The proposed framework is designed around the principle that vision-based motion analysis should be organized according to engineering functionality rather than individual computer vision algorithms. Instead of constructing a fixed computational pipeline, the framework separates the complete motion analysis process into independent engineering tasks connected through standardized geometric representations. This organization enables alternative computational methods to be incorporated without modifying downstream processing modules, thereby improving flexibility, extensibility, and software maintainability.

The overall architecture of the proposed framework is illustrated in **Figure 1**.

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              Proposed Framework

        ┌──────────────────────┐
        │ Feature Tracking      │
        └─────────┬────────────┘
                  │
        Feature Observations
                  │
                  ▼
     ┌──────────────────────────┐
     │ Coordinate Acquisition    │
     │                           │
     │ Measurement               │
     │ CAD                       │
     │ Stereo                    │
     └──────────┬───────────────┘
                │
     Object-space Coordinates
                │
                ▼
     ┌──────────────────────────┐
     │ Pose Estimation           │
     │ SolvePnP / Kabsch         │
     └──────────┬───────────────┘
                │
        Rigid-body Pose
                │
                ▼
  Motion Analysis
        │
  Signal Processing
        │
  Visualization
```

Figure 1. Overall architecture of the proposed modular computer vision framework.

The framework consists of six principal engineering components: image acquisition and feature tracking, object-space coordinate generation, pose estimation, motion analysis, signal processing, and visualization. Each component communicates exclusively through standardized engineering data structures representing feature observations, object-space coordinates, rigid-body poses, and motion histories. Consequently, computational modules remain independent of the specific algorithms employed by neighbouring processing stages.

This architectural separation permits multiple computational workflows to coexist within the same software environment. For example, object-space coordinates may originate from direct engineering measurements, CAD-assisted extraction, or stereo reconstruction, while rigid-body pose may subsequently be estimated using either monocular image observations or reconstructed three-dimensional geometry.

Regardless of the selected workflow, downstream engineering modules operate upon identical pose and motion representations.

3.2 Unified Engineering Representation

A central feature of the proposed framework is the adoption of a unified engineering representation that standardizes the exchange of geometric information throughout the computational workflow. Rather than allowing individual algorithms to define their own internal data structures, all engineering modules communicate through common representations of feature observations, object-space coordinates, rigid-body poses, and motion histories.

This abstraction separates engineering functionality from algorithm-specific implementation. Consequently, the motion analysis, signal processing, visualization, and reporting modules remain independent of the computational methodology used to acquire geometric information or estimate rigid-body pose. The resulting architecture improves software extensibility while reducing dependencies between individual computational components.

3.3 Modular Processing Workflows

The framework supports multiple computational pathways for performing equivalent engineering tasks. Three alternative workflows are currently implemented for object-space coordinate generation: direct engineering measurement, CAD-assisted coordinate extraction, and stereo reconstruction. Similarly, rigid-body pose estimation may be performed using monocular image observations through the **Perspective-n-Point (PnP)** problem, implemented using OpenCV's SolvePnP algorithm **[10,11,13]**, or through rigid-body registration of reconstructed three-dimensional feature coordinates using the **Kabsch algorithm [16]**.

Rather than treating these methodologies as competing algorithms, the framework considers them interchangeable computational realizations of common engineering functions. Once standardized engineering representations have been generated, all subsequent processing stages—including motion analysis, signal processing, visualization, and reporting—operate identically regardless of the selected computational workflow.

This modular organization allows additional coordinate acquisition or pose estimation techniques to be incorporated in the future without modification of the downstream engineering software.

3.4 Framework Operation

The complete engineering workflow implemented within the proposed framework is illustrated in **Figure 2**.

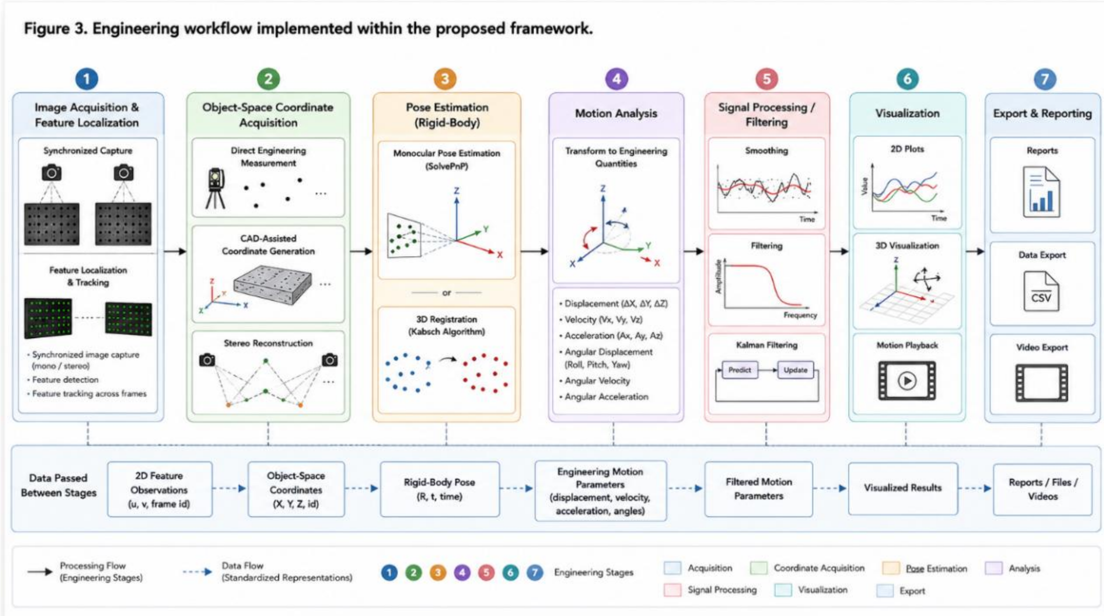

Figure 2. Engineering workflow implemented within the proposed framework.

The workflow begins with synchronized image acquisition and feature localization. Object-space coordinates are subsequently obtained using one of the supported coordinate acquisition strategies before being supplied to the selected pose estimation workflow. The resulting rigid-body poses are transformed into engineering motion parameters, optionally refined through configurable signal processing techniques, and finally presented using synchronized visualization and reporting tools.

By organizing the complete measurement process into modular engineering stages connected through standardized geometric representations, the proposed framework accommodates diverse measurement scenarios while preserving a consistent computational architecture. This organization represents the principal contribution of the proposed work and distinguishes the framework from conventional software systems based upon fixed algorithmic processing pipelines.

4. Software Realization

The modular framework presented in the previous section was realized as an integrated desktop software system supporting the complete workflow of vision-based motion analysis. Rather than functioning as a collection of independent computer vision utilities, the software was designed as a unified engineering environment in which image acquisition, geometric processing, motion analysis, visualization, and reporting are performed within a single application. Throughout the implementation, the modular architecture and standardized engineering data representations proposed in Section 3 were preserved, allowing individual computational components to operate independently while exchanging information through common geometric structures.

The implementation was guided by three principal design objectives. First, each engineering task should be implemented as an independent software module with clearly defined responsibilities. Second, computational algorithms should remain interchangeable without affecting downstream processing stages. Finally, the software should provide engineers with an integrated workflow that supports complete motion analysis, from image acquisition to quantitative reporting, without requiring multiple external applications.

4.1 Software Architecture

The implemented software follows the modular organization proposed in Section 3, with each engineering task realized as an independent software component. Individual modules communicate through standardized engineering data structures representing feature observations, object-space coordinates, rigid-body poses, and motion histories. Consequently, modifications within one computational stage do not require changes to subsequent stages, allowing alternative computer vision algorithms to be incorporated while preserving the overall software architecture.

The principal software modules implemented within the framework are illustrated in **Figure 3**.

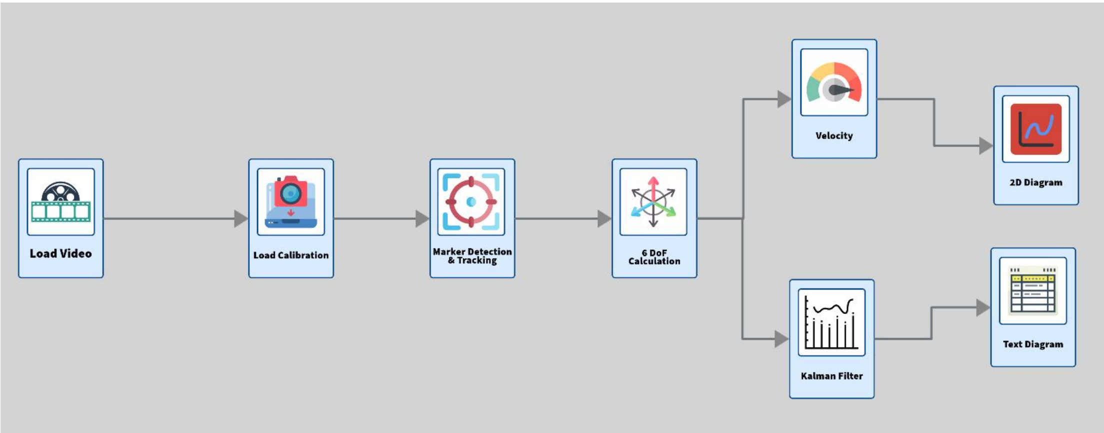

Figure 3. Principal software module implementing the proposed engineering framework.

After the figure:

As shown in Figure 3, the software begins with image acquisition and calibration loading before proceeding through feature tracking, coordinate generation, pose estimation, motion analysis, visualization, and data export. This organization closely follows the engineering workflow proposed by the framework while maintaining clear separation between individual computational responsibilities.

4.2 Technology Stack

The proposed framework was implemented using a combination of established software libraries selected to support computer vision, numerical computation, three-dimensional visualization, and desktop application development. Rather than developing new computational algorithms, the implementation integrates mature open-source technologies within a unified engineering architecture.

The software was developed using C# and the Windows Presentation Foundation (WPF) framework, providing a flexible desktop environment for interactive engineering analysis. Computer vision functionality was implemented through **OpenCvSharp [19]**, enabling seamless integration of **OpenCV [13]** algorithms for camera calibration, feature detection and tracking, stereo reconstruction, and monocular pose estimation within the C# application. Three-dimensional visualization of reconstructed geometry and engineering coordinate systems was realized using **HelixToolkit [20]**, while numerical computations involving rigid-body transformations and matrix operations were performed using **Math.NET Numerics [21]**.

The combination of these technologies enabled the implementation of a modular engineering application while preserving compatibility with widely adopted computer vision and numerical computation libraries.

Table 1. Software Technologies Used

Technology	Purpose
C#	Core application development
WPF	Graphical user interface
OpenCvSharp	Computer vision algorithms
HelixToolkit	Three-dimensional visualization
Math.NET Numerics	Matrix operations and numerical computation

4. 3 Core Functional Modules

4.3.1 Camera Calibration Module

The camera calibration module establishes the geometric relationship between image observations and object space, providing the intrinsic and extrinsic parameters required for subsequent three-dimensional reconstruction and pose estimation. The implementation supports both intrinsic and stereo calibration workflows while maintaining compatibility with the modular framework architecture.

Intrinsic calibration is implemented using OpenCV through the OpenCvSharp wrapper **[13,19]**. Calibration target corners are localized using **findChessboardCornersSB**, after which **calibrateCamera**, based on Zhang's calibration methodology **[5]**, estimates the intrinsic camera matrix and lens distortion coefficients. Stereo calibration is subsequently performed using **stereoCalibrate** and **stereoRectify [13]** to recover the relative camera geometry required for stereo reconstruction.

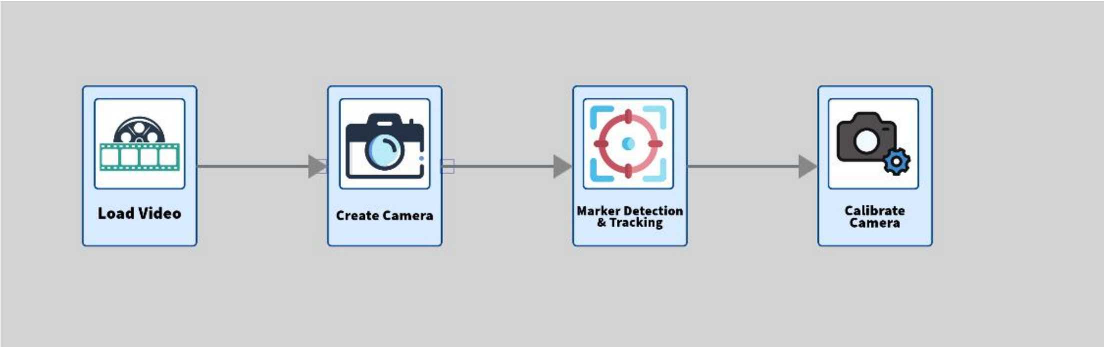

Figure 4. Camera intrinsic calibration module.

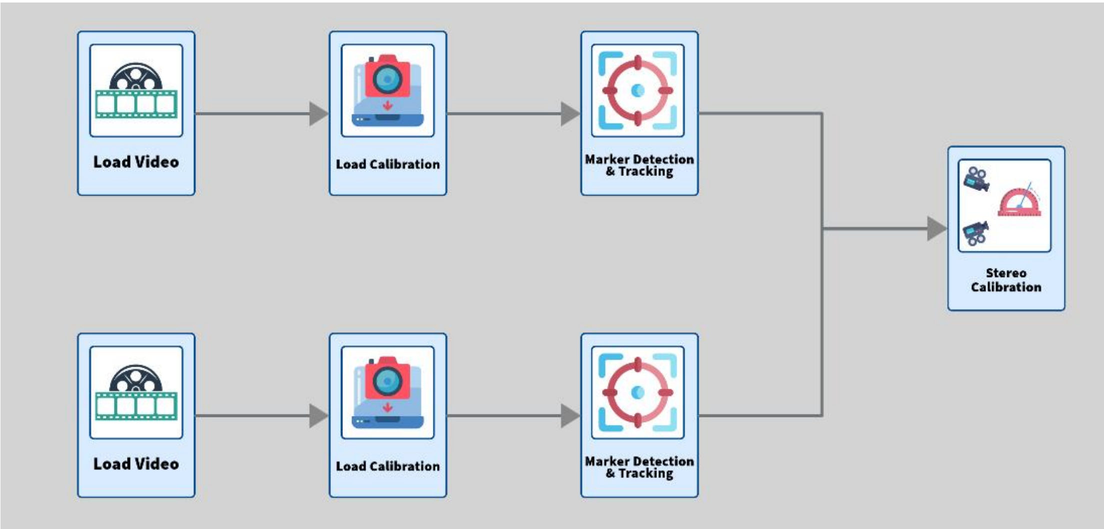

Figure 5. Stereo calibration module for 2 camera stereo system.

The resulting calibration parameters are stored within the standardized engineering data model and are subsequently utilized by the coordinate generation and pose estimation modules without exposing algorithm-specific implementation details to downstream components.

4.3.2 Feature Tracking Module

The feature tracking module is responsible for maintaining temporal correspondence between engineering features throughout an image sequence. Unlike conventional tracking systems that apply a single tracking methodology globally, the proposed implementation permits the tracking strategy to be assigned independently to each feature according to its geometric characteristics and engineering requirements.

The implemented tracking methods include **Lucas–Kanade optical flow [8]**, **CSRT-based tracking [15]**, correlation tracking, quadrant symmetry, circular symmetry, template-based localization, and manual correction procedures. In addition, the framework supports virtual feature generation through geometric relationships between previously tracked physical features, allowing engineering reference points to be maintained even when no corresponding physical image feature exists.

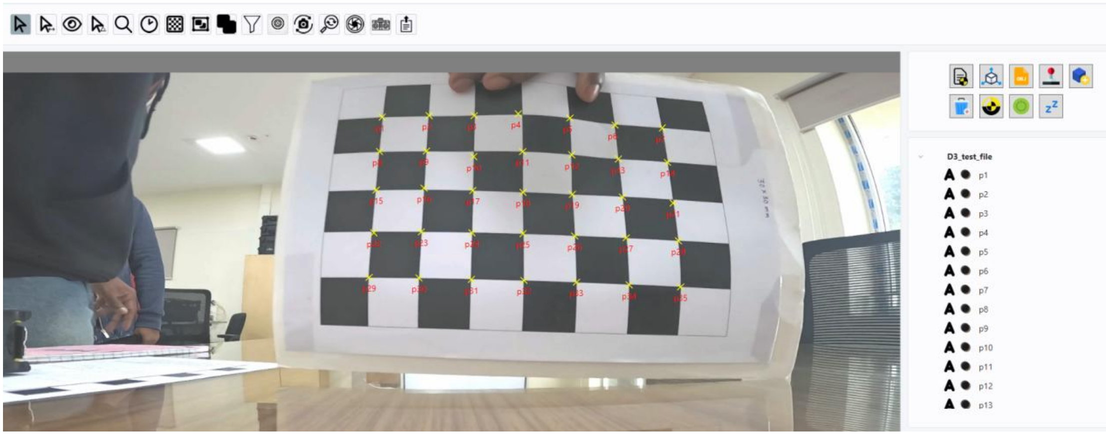

Figure 6. Feature Tracking with OpenCv algorithms.

Feature observations are preserved as complete temporal histories rather than isolated frame measurements, providing the standardized engineering representations required by the coordinate generation, pose estimation, and motion analysis modules.

4.3.3 Object-Space Coordinate Generation Module

The coordinate generation module transforms tracked image observations into standardized object-space representations suitable for engineering computation. Rather than assuming a single source of geometric information, the implementation supports multiple coordinate acquisition workflows, including direct engineering measurement, CAD-assisted coordinate extraction, and stereo reconstruction.

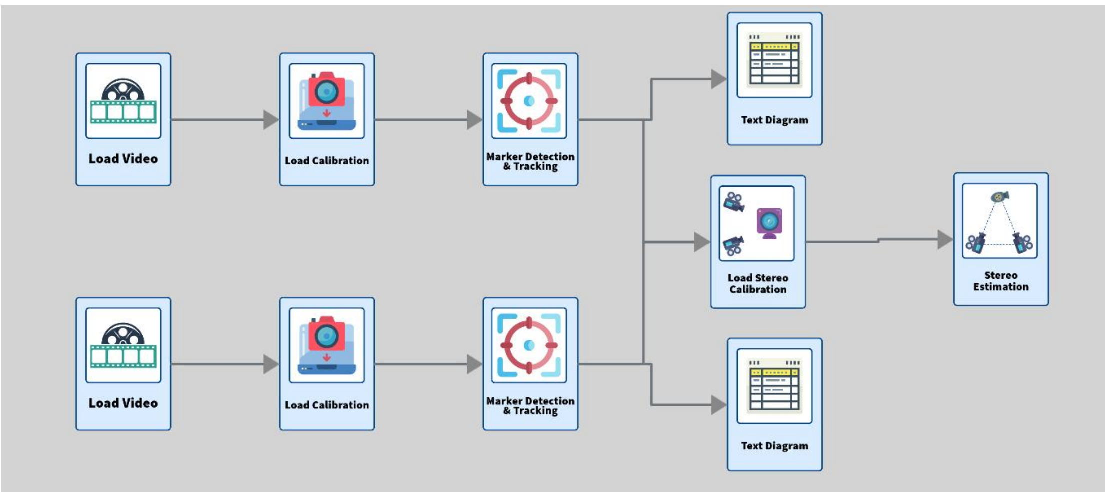

Figure 7. Object space coordinate generation module via stereo coordinate estimation.

Stereo reconstruction is implemented using calibrated stereo camera geometry together with OpenCV triangulation procedures **[2,13,14]**, while CAD-assisted workflows utilize imported engineering models displayed through the integrated three-dimensional visualization environment. Regardless of the selected acquisition methodology, all generated coordinates are converted into a common engineering representation before being supplied to the pose estimation module.

4.3.4 Pose Estimation Module

The pose estimation module determines the rigid-body transformation describing the position and orientation of the observed object. Consistent with the modular philosophy of the framework, multiple computational methodologies are supported while producing identical engineering outputs.

Monocular pose estimation is implemented using OpenCV's SolvePnP algorithm **[10,11,13]**, with orientation represented through rotation vectors converted using the Rodrigues transformation **[11]**. Stereo-based pose estimation is implemented through rigid-body registration using the Kabsch algorithm **[16]** operating directly on reconstructed three-dimensional feature coordinates. Matrix operations required for rigid-body registration are performed using Math.NET Numerics [21].

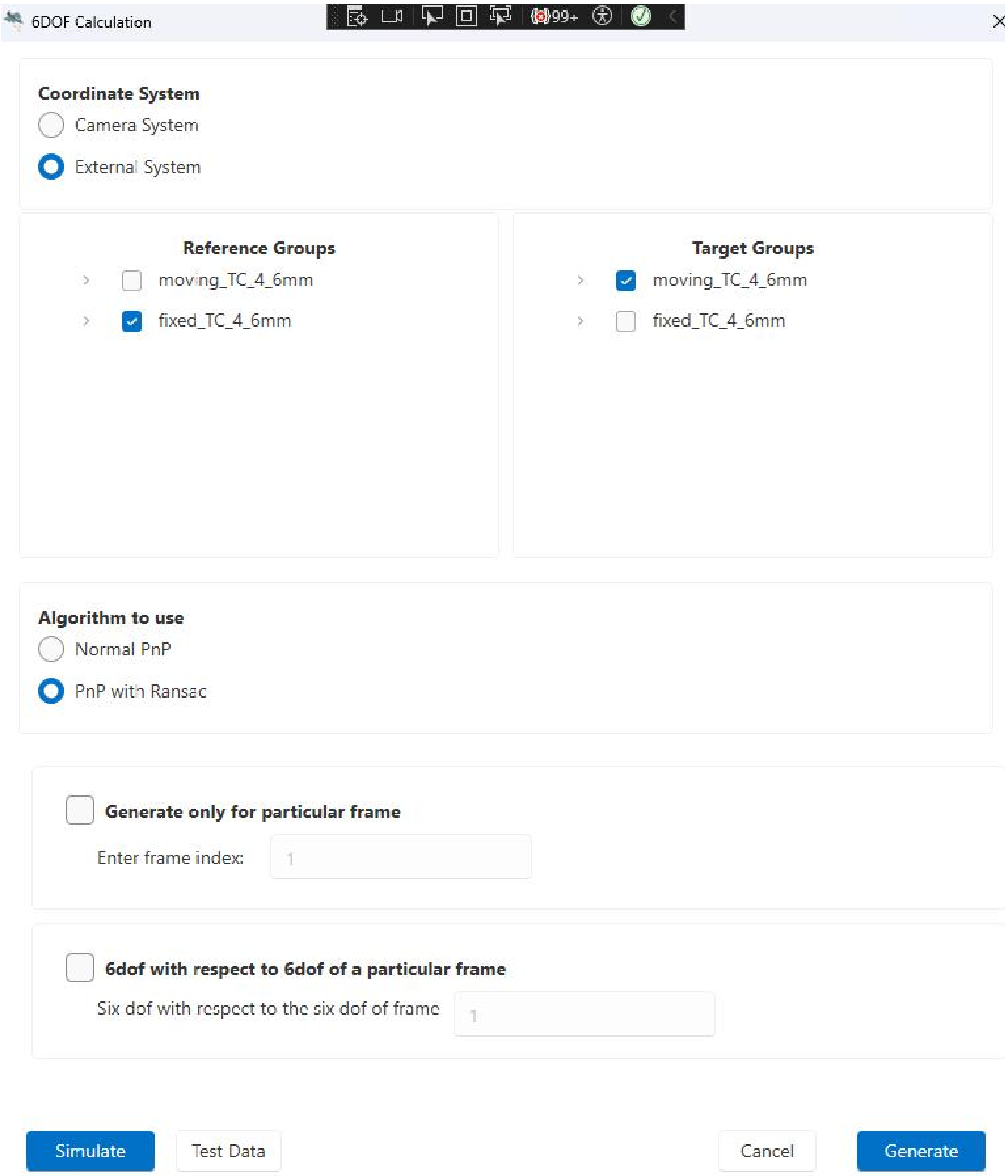

Figure 8. Representative monocular pose estimation using the implemented SolvePnP workflow.

Both workflows generate standardized rigid-body pose representations that are subsequently processed identically by the motion analysis module.

4.3.5 Motion Analysis Module

The motion analysis module transforms successive rigid-body poses into quantitative engineering measurements describing translational and rotational motion. Motion quantities including displacement, velocity, acceleration, angular velocity, and angular acceleration are computed from synchronized pose histories maintained throughout the measurement sequence.

To improve measurement stability, the framework incorporates a configurable signal processing environment supporting Moving Average, FIR, IIR, CFC **[22]**, Savitzky–Golay **[23]**, and Kalman filtering **[24]**.. Since filtering operates exclusively upon standardized engineering motion histories, alternative signal processing methodologies may be selected without affecting the remaining computational workflow.

4.3.6 Visualization and Reporting Module

The visualization module provides synchronized presentation of image observations, reconstructed three-dimensional geometry, rigid-body pose, engineering motion histories, and numerical analysis results within a unified software environment. Interactive visualization is implemented using WPF together with **HelixToolkit [20]** for three-dimensional rendering and integrated plotting controls for motion analysis.

The software further supports engineering reporting through graphical plots, tabulated measurement data, and export of processed results, enabling the complete computational workflow to be performed within a single application.

4.4 Software Operation

The implemented software provides a unified workflow for conducting complete vision-based motion analysis. A typical engineering investigation begins by loading the recorded image sequence together with the corresponding camera calibration parameters. Feature observations are subsequently obtained through the integrated tracking module before object-space coordinates are generated using the selected coordinate acquisition strategy.

The estimated geometric information is then supplied to the appropriate pose estimation workflow, producing rigid-body transformations that form the basis for subsequent motion analysis. Engineering motion parameters are computed directly from the resulting pose histories and may be further refined using configurable filtering techniques before being presented through synchronized graphical visualization, three-dimensional rendering, and engineering reports.

By integrating the complete processing sequence within a single software environment, the framework eliminates the need for transferring intermediate data between multiple applications while maintaining the flexibility to accommodate alternative computational methodologies through its modular architecture.

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5. Experimental Validation

5.1 Experimental Setup

The proposed framework was evaluated using a series of real engineering measurement experiments designed to validate the principal computational stages of the motion analysis workflow. The experimental investigations included intrinsic and stereo camera calibration, object-space coordinate generation, monocular pose estimation, and engineering motion analysis. Together, these experiments evaluate both the numerical correctness of the implemented computational modules and the practical applicability of the proposed software architecture.

All experiments were performed using two Logitech C920 HD Pro cameras operating at a resolution of **1920 × 1080 pixels**. Intrinsic camera calibration was performed independently for each camera using a **5 × 8** checkerboard calibration target with a square size of **90 mm**. Calibration images were extracted from video sequences containing more than **2000 frames**, with every fiftieth frame selected for calibration to ensure sufficient variation in target orientation and position.

Stereo calibration was subsequently performed using the calibrated camera pair. Object-space coordinate generation and pose estimation experiments utilized representative engineering measurement scenarios in which independently measured reference data were available for validation. Motion analysis experiments were conducted using the estimated rigid-body poses generated by the framework.

Table 2 summarizes the principal experimental configuration.

Table 2. Experimental Configuration

Parameter	Value
Cameras	Logitech C920 HD Pro (2)
Image Resolution	1920 × 1080 pixels
Calibration Pattern	5 × 8 Checkerboard
Square Size	90 mm
Corner Detection	OpenCV findChessboardCornersSB**[13]**
Calibration Images	>2000 video frames (every 50th frame used)
Pose Estimation	OpenCV SolvePnP**[13]**
Stereo Registration	Kabsch Algorithm**[16]**

5.2 Camera Calibration

The intrinsic calibration procedure was performed independently for each camera using the implemented calibration workflow. The estimated intrinsic parameters exhibited consistent focal lengths, principal point locations, and lens distortion coefficients appropriate for the employed imaging configuration. The resulting reprojection errors remained below one pixel for both cameras, indicating satisfactory calibration accuracy for engineering measurement applications.

The intrinsic calibration parameters obtained for the camera is summarized in **Figure 9**, while representative calibration image is shown in **Figure 10**.

Figure 9. Intrinsic calibration parameters estimated for a representative camera.

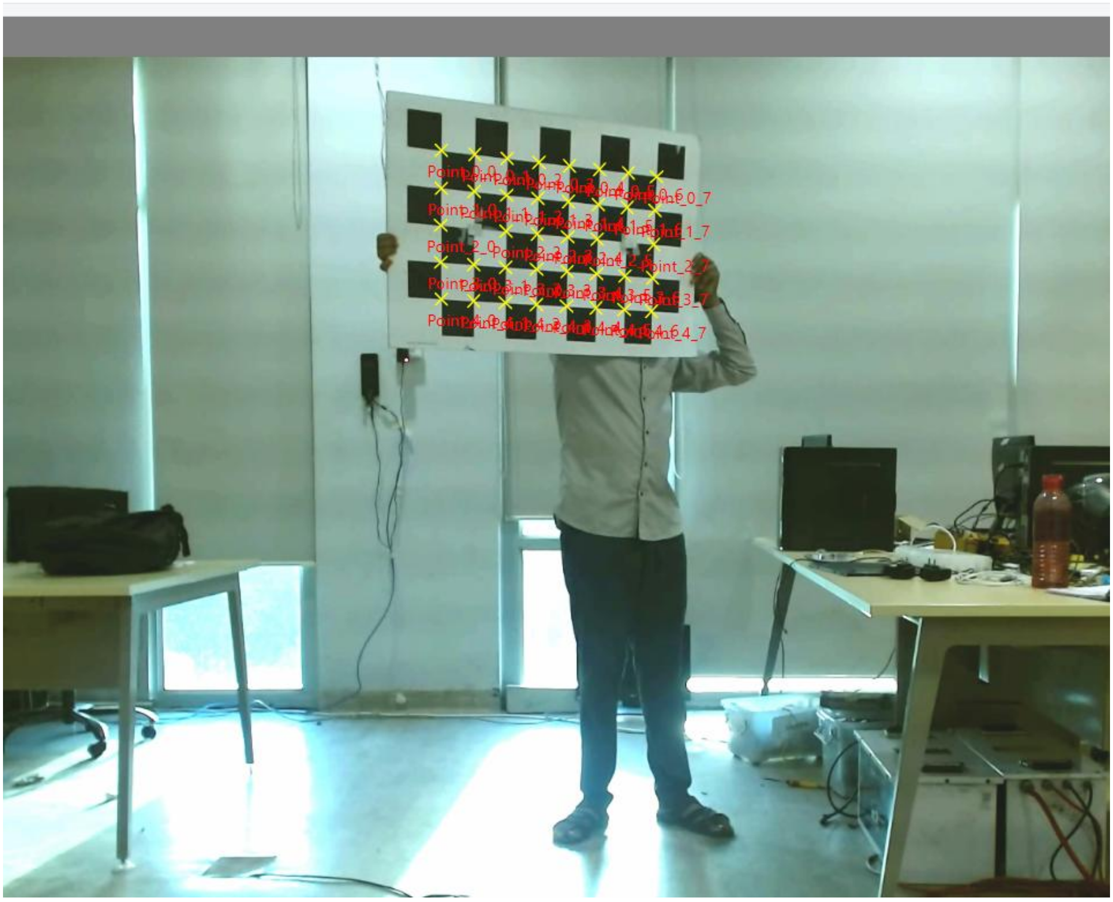

Figure 10. Representative checkerboard image used for intrinsic calibration.

The low reprojection errors obtained for both cameras indicate that the estimated intrinsic parameters provide an accurate geometric model of the imaging system. Since camera calibration establishes the geometric foundation for stereo reconstruction and pose estimation, these results demonstrate that the calibration module provides sufficiently accurate parameters for subsequent engineering measurements performed within the proposed framework.

5.3 Stereo Calibration

Following intrinsic calibration, stereo calibration was performed to determine the relative orientation and translation between the two cameras. Two independent calibration trials were conducted using representative image sets. The recovered baseline remained consistent with the physical camera arrangement, while the estimated relative orientation provided the geometric relationship required for stereo reconstruction.

The stereo calibration results are summarized in **Table 3**.

Table 3. Stereo Calibration Results

Trial	RMS Error	Estimated Baseline
1	1.139	0.376 m
2	0.785	0.712 m

The recovered stereo geometry demonstrated consistent estimation of the relative camera orientation and baseline, providing the geometric relationship required for accurate stereo triangulation. These results validate the implementation of the stereo calibration module and establish the foundation for reliable object-space coordinate generation within the proposed framework.

5.4 Object-Space Coordinate Generation

The proposed framework supports multiple methods for obtaining object-space coordinates, including direct engineering measurement, CAD-assisted extraction, and stereo reconstruction. Experimental validation focused primarily on the stereo reconstruction workflow because it represents the most computationally demanding coordinate acquisition strategy.

Stereo reconstruction was evaluated using synthetic datasets possessing known geometric ground truth, enabling direct comparison between reconstructed and reference object-space coordinates. The reconstruction procedure employed calibrated stereo geometry and triangulation methods based on established multiple-view geometry principles **[2,14]**. The recovered camera geometry and triangulated coordinates demonstrated accurate reconstruction across the evaluated datasets, confirming the correctness of the implemented stereo workflow.

Representative stereo image pairs employed during the reconstruction experiments are shown in **Figure 11.**

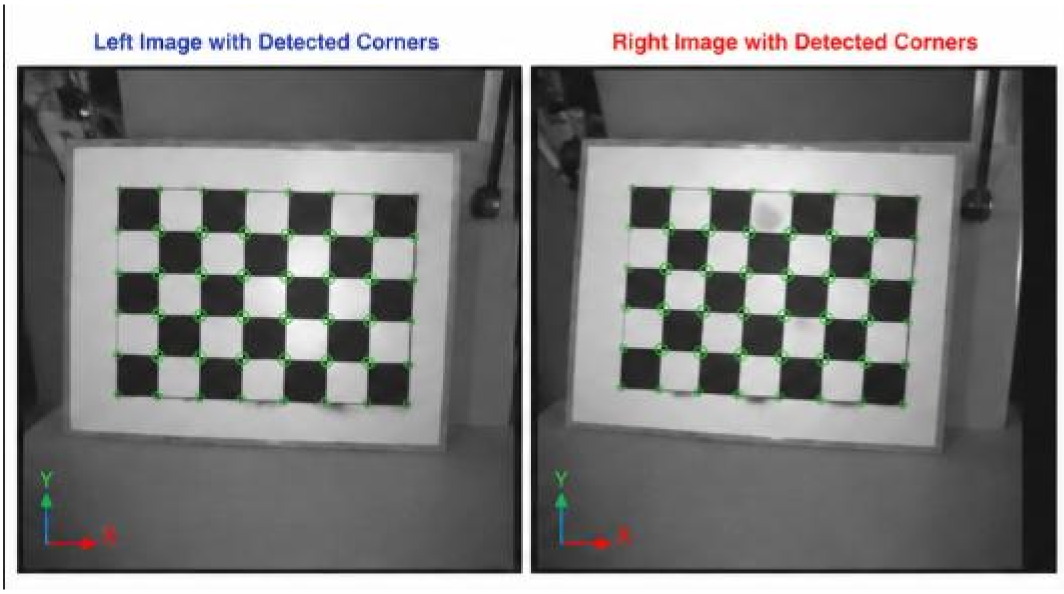

Figure 11. Representative synthetic stereo image pair used for reconstruction validation.

The reconstruction experiments confirmed that the implemented stereo workflow accurately recovered object-space coordinates from synchronized image observations while preserving compatibility with the standardized engineering data representation employed throughout the framework. The successful reconstruction of known synthetic geometry demonstrates that the coordinate generation module provides reliable three-dimensional measurements suitable for subsequent pose estimation and motion analysis.

5.5 Monocular Pose Estimation

The monocular pose estimation workflow was evaluated using independently obtained reference pose measurements derived from total-station coordinate measurements and rigid-body registration using the Kabsch rigid-body registration **[16]**. The estimated pose obtained through the implemented SolvePnP workflow **[13]** was compared directly with the corresponding reference transformations.

Table 4 illustrates representative comparisons between the estimated and reference pose histories, while figure12 presents the corresponding reprojection obtained using the estimated rigid-body transformation.

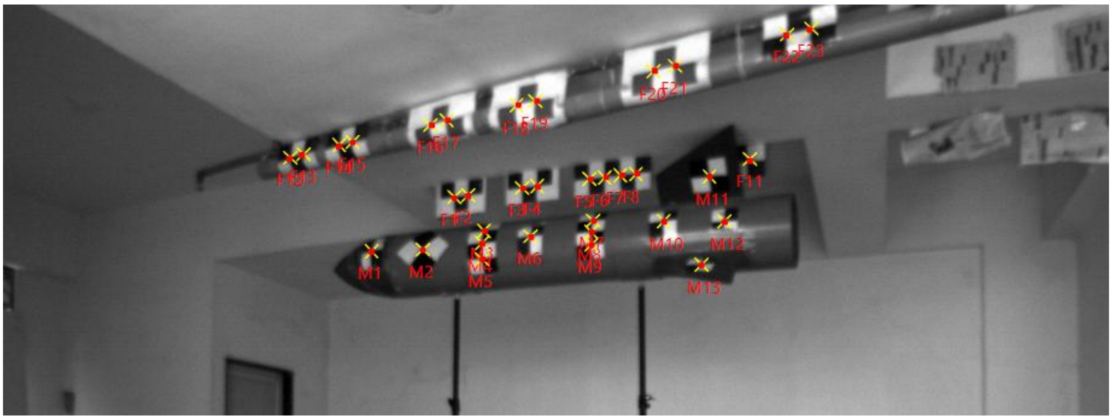

Figure 12. Reprojected feature points obtained using the estimated SolvePnP pose.

The quantitative pose estimation accuracy is summarized in **Table 4**.

Table 4. Pose Estimation Accuracy

Quantity	Mean Absolute Error	Maximum Error	RMSE
X (m)	0.0415	0.1206	0.0582
Y (m)	0.0150	0.0387	0.0182
Z (m)	0.0107	0.0301	0.0139
Roll (deg)	0.6379	1.5596	0.7989
Pitch (deg)	0.3571	0.9287	0.4556
Yaw (deg)	0.4187	1.1565	0.5169

The observed translational and rotational errors demonstrate that the implemented monocular pose estimation workflow provides reliable input for subsequent engineering motion analysis.

The estimated rigid-body pose exhibited good agreement with the independently determined reference measurements obtained using total-station observations and rigid-body registration. The relatively small translational and rotational errors demonstrate that the implemented SolvePnP workflow provides reliable rigid-body pose estimates under practical engineering conditions. These results further confirm that the standardized object-space coordinate representations generated by the framework can be successfully utilized for accurate vision-based pose estimation.

5.6 Motion Analysis

The estimated rigid-body poses were subsequently transformed into engineering motion quantities, including translational displacement, rotational displacement, linear velocity, angular velocity, and acceleration. Configurable signal processing techniques, including CFC **[22]**, Savitzky–Golay **[23]**, and Kalman filtering **[24]**, were applied to reduce measurement fluctuations while preserving the underlying motion characteristics.

Representative motion histories generated by the framework are presented in **Figure 13**, while **Figure 14** illustrates the effect of the implemented filtering procedure on the computed engineering measurements.

Velocity1-Frame	Velocity1-Time (s)	Velocity1-X (unit/s)	Velocity1-Y (unit/s)	Velocity1-Z (unit/s)	Velocity1-Roll (rad/s)	Velocity1-Pitch (rad/s)	Velocity1-Yaw (rad/s)
0	0	NaN	NaN	NaN	0.43529764865068593	-0.1207035464153518	0.28682644266990787
1	1	0.19382825354330446	0.0010386462706694077	-0.003472983898096249	0.20215115934770103	-0.03659991402382627	0.51009508559879
2	2	0.13961189509713723	0.02186067264620062	-0.012635352588256876	-0.651260800296178	0.11977982584500127	-0.3004099671318994
3	3	0.20773858405882084	0.004753040327252767	-0.005333901488208226	-1.132363465116435	-0.4963283850884662	-0.3122167890691085
4	4	0.32010729093519164	-0.021387709539731903	0.01759554774793093	0.4209372080056861	-0.16597419600850777	-0.046710911327614846
5	5	0.28345387656128995	-0.0007960022482084161	0.009306904802066085	0.6921389460486119	0.4991230369829893	0.04706565775470001
6	6	0.25378694316288986	0.01540195548671508	-0.00526656311078999	0.05137724365243701	0.19856245647746656	0.24820173222618683
7	7	0.31873544549150723	-0.003552747402650769	0.005147239428192951	0.5772470937130467	0.29187389670082237	0.5232829161366135
8	8	0.4325812388767274	-0.015668416974825683	0.01795598726007852	-0.18571809032970338	0.5810785665716707	-0.013384610231957472
9	9	0.6991693956169083	0.03554205371784813	-0.004787557761168348	3.7374025895132874	1.4498219989845031	-1.5235963693064327
10	10	2.888000156280797	-1.9028706874853136	-4.892440105195609	60.78641043279045	21.30052667000703	-0.41404089356801754
11	11	NaN	NaN	NaN	NaN	NaN	NaN

Figure 13. Representative translational and rotational motion histories generated by the framework.

Kalmanfilter1-Frame	Kalmanfilter1-Time (s)	Kalmanfilter1-X (mm)	Kalmanfilter1-Y (mm)	Kalmanfilter1-Z (mm)	Kalmanfilter1-Roll (rad)	Kalmanfilter1-Pitch (rad)	Kalmanfilter1-Yaw (rad)
0	0	1.0216393377289084	-0.3646934728203237	0.4624846153287301	0.06586056760139936	0.016345412278893093	-0.20201972025991452
1	1	1.4702851181500534	-0.4809605088212093	0.5966572688191437	0.7049568330712722	-0.15091369962987172	0.14849447031833501
2	2	1.6028911801163068	-0.44973916124620616	0.5625610828859049	0.5995552477015068	-0.09395917134345898	0.6717099512509137
3	3	1.6385930059569327	-0.3875497721883303	0.4986644666388783	-0.12758860761115554	-0.010767812339883942	0.03062798220839147
4	4	1.8394145629026744	-0.36703116927997986	0.4662980263910745	-1.4947128160959453	-0.8139464551480158	0.12032526339584554
5	5	2.1383558406983183	-0.3779938433004336	0.47378725608195865	-0.13403048622690372	-0.5074990840805634	-0.23549151515379046
6	6	2.393667652634334	-0.36151019862692807	0.46729156242740366	-0.29635241953048563	-0.15996943536027225	0.1305686847842018
7	7	2.662994003658655	-0.3507635021622108	0.4643975466854816	0.3974655276475629	0.05978809089803956	0.18144861910823254
8	8	3.006833014907525	-0.35779737438937337	0.47147287242488806	0.7476402068611232	0.4692632075575832	1.096676381860376
9	9	3.482719599053805	-0.37525521487401653	0.4938974273901174	0.4031195144008042	1.1372170276012294	0.4594506617515901
10	10	4.305709544360775	-0.3110231184745018	0.47340439387326344	6.5142826591732375	3.0348005587488016	-1.2524408826280562
11	11	8.34537424144366	-3.3333495313588464	-7.153936425941095	97.24752112944496	35.18187538453385	-0.9959164024241827

Figure 14. Comparison between raw and filtered engineering motion measurements.

These investigations demonstrate that the proposed framework successfully integrates calibration, coordinate generation, pose estimation, signal processing, and visualization into a complete engineering workflow suitable for quantitative vision-based motion analysis.

The motion analysis experiments demonstrate that the proposed framework successfully transforms validated rigid-body pose estimates into meaningful engineering motion quantities suitable for quantitative analysis. Furthermore, the integration of configurable signal processing techniques improves measurement stability while preserving the underlying motion characteristics. Collectively, these results confirm that the complete computational workflow—from camera calibration through visualization—operates as an integrated engineering system capable of supporting practical vision-based motion analysis.

5.7 Experimental Summary

The experimental investigations validate each principal stage of the proposed framework, including camera calibration, stereo geometry estimation, object-space coordinate generation, monocular pose estimation, and motion analysis. The results demonstrate that the implemented software provides reliable geometric measurements while maintaining the modular engineering architecture proposed in this work. Together, the experiments confirm that established computer vision algorithms can be integrated into a unified software framework capable of supporting flexible and quantitative vision-based motion analysis.

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6. Discussion

The experimental investigations demonstrate that the proposed framework successfully integrates established computer vision methodologies within a unified engineering architecture for vision-based motion analysis. Rather than introducing new computational algorithms, the framework focuses on the organization and integration of existing techniques into a modular software environment capable of supporting multiple engineering workflows through standardized geometric representations.

The calibration experiments confirmed that the implemented intrinsic and stereo calibration procedures provide reliable geometric models for subsequent three-dimensional reconstruction and pose estimation. The recovered camera parameters and stereo geometry enabled consistent object-space coordinate generation, establishing the geometric foundation required for quantitative engineering measurements.

The pose estimation experiments further demonstrated that the implemented monocular workflow produced rigid-body transformations that closely agreed with independently generated reference measurements obtained from total-station observations. These results indicate that the framework is capable of providing reliable pose estimates suitable for practical engineering motion analysis. Furthermore, the successful integration of a stereo-based pose estimation workflow illustrates the architectural flexibility of the proposed framework by enabling alternative computational approaches to coexist within the same software environment.

One of the principal contributions of the proposed framework lies in the separation of engineering functionality from algorithm-specific implementation. By representing feature observations, object-space coordinates, rigid-body poses, and motion histories using standardized engineering data structures, downstream computational modules remain independent of the methods used for coordinate acquisition or pose estimation. Consequently, new computational techniques may be incorporated into the framework with minimal impact on the remaining software architecture.

The integrated software environment additionally demonstrates the practical advantages of combining image processing, geometric computation, motion analysis, signal processing, visualization, and reporting within a single engineering platform. Instead of requiring multiple independent software tools, the proposed framework provides a unified environment supporting the complete workflow from image acquisition to engineering interpretation.

Although the present implementation currently incorporates direct engineering measurement, CAD-assisted coordinate generation, stereo reconstruction, SolvePnP, and Kabsch registration, the modular organization permits additional methodologies to be integrated without modification of the overall computational architecture. This extensibility makes the framework applicable to a broad range of engineering measurement scenarios while facilitating future development as new computer vision techniques become available.

The experimental results support the central architectural principle proposed in this work: that vision-based motion analysis can be organized around engineering functionality rather than individual computational algorithms. Despite employing established computer vision techniques for calibration, reconstruction, and pose estimation, the standardized engineering representations adopted by the framework enabled multiple computational workflows to operate interchangeably without modification of downstream processing modules. This demonstrates that the proposed architecture provides flexibility at the software level while preserving the numerical reliability of the underlying computer vision methods.

Overall, the experimental results demonstrate that the proposed framework achieves its primary objective of providing a flexible and extensible architecture for vision-based motion analysis. The work therefore contributes not through the development of new computer vision algorithms, but through the systematic integration of established methodologies into a coherent engineering framework capable of supporting diverse measurement workflows within a common computational environment.

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7. Conclusion

This paper presented a modular computer vision framework for flexible vision-based motion analysis based on common geometric representations. Unlike conventional software systems that organize processing around fixed computational pipelines, the proposed framework separates engineering functionality from algorithm-specific implementation, allowing multiple coordinate acquisition and pose estimation methodologies to operate within a unified computational architecture.

The framework integrates established computer vision techniques, including camera calibration, feature tracking, object-space coordinate generation, monocular and stereo-based pose estimation, motion analysis, signal processing, visualization, and engineering reporting into a coherent software environment. By expressing geometric information through standardized engineering representations, downstream computational modules remain independent of the specific algorithms employed during preceding processing stages, thereby improving software flexibility, extensibility, and maintainability.

Experimental validation using real engineering measurement data demonstrated the practical applicability of the proposed framework. The calibration experiments produced reliable intrinsic and stereo camera models suitable for quantitative measurement, while pose estimation experiments showed good agreement between the estimated rigid-body transformations and independently obtained reference measurements. The resulting motion analysis workflow successfully transformed image observations into quantitative engineering measurements suitable for subsequent interpretation and analysis.

The principal contribution of this work is not the introduction of new computer vision algorithms, but the development and validation of an engineering framework that systematically integrates established computational methodologies within a modular software architecture. This organization enables diverse engineering measurement workflows to be realized using common geometric representations while facilitating future incorporation of additional computer vision techniques without requiring modification of the overall computational workflow.

Future work will focus on extending the framework through the integration of additional feature tracking methodologies, alternative three-dimensional reconstruction techniques, advanced pose estimation algorithms, automated feature detection, and real-time processing capabilities. These developments will further improve the adaptability of the framework to a broader range of engineering motion analysis applications while preserving the modular architectural principles established in this work.

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